Geometry
Unit 9
Lesson 2 Practice

Name____Answer Key__
Date_________________
Period________________

1. Pygmy Date Palm Trees are common in landscaping in Florida. Herbert bought 12 trees for his yard. Each palm tree needs approximately 16 square feet for its root system to grow. How many square feet does Herbert need of his new trees?

$12 \times 16 = 192$ ft^2

2. Herbert plans to plant a line of trees along his property line. The area he decided upon is 200 square feet. Does he have sufficient space for his 12 new trees?

Yes, 200 is more than 192.

3. If Herbert wanted to spread out the trees so that each palm tree had approximately 25 square feet, would he still have sufficient space for all 12 trees?

$25 \times 12 = 300$ sq. ft
No, 300 is more than 200.

4. What is the density of palm trees if Herbert allows only 16 square feet for each palm tree?

$\frac{1 \text{ tree}}{16 \text{ sq. ft}} = 0.0625$ tree/sq. ft

5. What is the density of palm trees if Herbert allows 25 square feet for each palm tree?

$\frac{1}{25} = 0.04$ tree/sq. ft

6. Lake George, located on the St. Johns River in Florida, covers approximately 71.88 square miles. In 2018, a survey noted there were $2{,}322$ alligators counted there. What is the density of alligators at Lake George?

$\frac{2322}{71.88} = 32.3$ alligators/mi^2

7. If 26 of the alligators counted were bull alligators, what is the density of bull alligators?

$\frac{26}{71.88} = 0.36$ bull alligators/mi^2

8. Walt Disney World spreads over 42.47 square miles. If $57{,}000$ guests enter Walt Disney World on any given day, what is the population density of visitors?

$\frac{57000}{42.47} = 1342.1$ guests/sq. mile

9. San Francisco, California covers 46.87 square miles. The population of San Francisco in 2019 was $874{,}961$. What was the population density of San Francisco in 2019?

$\frac{874961}{46.87} = 18{,}668$ people/sq. mile

10. Walt Disney World is close to the same size as San Francisco. How do the population densities compare, assuming a full day of visitors were at Walt Disney World?

Answers will vary.
Sample answers:

- Walt Disney World is denser than San Francisco.
- $\frac{18668}{1342.1} \approx 14$; Walt Disney World is about 14 times denser in San Francisco.
- $\frac{1342.1}{18668} \approx \frac{7}{10}$; San Francisco is about $\frac{7}{10}$ as dense as Walt Disney World